

Active Learning Guide

Simulator: Graphical Method – Linear Programming (2 variables)

1. Introducción

The simulator allows the user to define a mathematical programming model in order to obtain its graphical representation and solve **Linear Programming problems with two decision variables**.

Its purpose is not merely to compute the optimal solution of a linear optimization problem, but to facilitate both geometric and conceptual understanding of:

- The feasible region.
- Binding constraints (i.e., bottleneck constraints) and/or redundant constraints.
- Basic Feasible Solutions (BFSs) and Basic Non-Feasible Solutions (BNFSs).
- The behavior of the objective function.
- The sensitivity of the solution to parametric variations.
- Special cases: infeasibility, unboundedness, and multiple optimal solutions.

This guide is designed to promote **active learning** through systematic experimentation with the model.

2. Functional Structure of the Simulator

The interface is divided into two main areas:

◆ **Model Definition Panel**, where the input parameters are specified. This panel contains the **Constraints** block, the **Objective Function**, and the **Nature of the Variables**, all of which can be configured parametrically.

◆ **Visualization and Results Panel**, which includes:

- Graphical representation of the feasible region.
- Visualization of the lines associated with the constraints.
- Automatic computation of the optimal solution.
- Complete algebraic formulation of the model.
- Status messages regarding the state of the problem.

Note: The design enables immediate visualization of the geometric effect of any modification to the model parameters.

3. Model Formulation

3.1 Linear Constraints

Each constraint takes the form:

$$a_{i1}x_1 + a_{i2}x_2 (\leq, \geq, =) b_i$$

Geometrically, each constraint of type “ $\leq$ ” or “ $\geq$ ” is represented as a half-space bounded by a straight line. The intersection of all half-spaces defines the **feasible region**, which in two dimensions is a convex polyhedron (a convex polygon). The simulator allows the user to:

- Configure the technological coefficients (a_{ij}) and the right-hand-side constants of the constraints (b_i)
- Add constraints (+).
- Delete the last constraint (-).
- Reset the entire set of constraints (↺).

Note: The title field associated with each constraint is purely descriptive and does not affect the computational procedure.

3.2 Objective function

The objective function is defined as:

$$z = c_1x_1 + c_2x_2$$

Through the simulator, it is possible to specify the optimization direction (maximize or minimize). Geometrically, the objective function generates a family of level curves (iso-cost or iso-profit lines) by assigning different values to the constant term k (corresponding to the cost or profit level):

$$c_1x_1 + c_2x_2 = k$$

When the optimum exists and is unique, it is attained at a **Basic Feasible Solution (BFS)**, that is, at a vertex (extreme point) of the feasible region.

3.3 Nature of the variables

By default, the following non-negativity constraints are imposed:

$$x_1 \geq 0, x_2 \geq 0$$

This restricts the analysis to the first quadrant of the plane. If these conditions are deactivated, the feasible domain may extend to other quadrants, substantially modifying the geometric structure of the problem.

3.4 Input of Coefficients and Tool Limitations

The simulator allows the user to enter the technological coefficients of the model (a_{ij}), the right-hand-side constants (b_i) and the objective function coefficients (c_j) using numerical values in decimal format.

Accepted Numerical Format

Coefficients must be entered in decimal format (e.g., 2,5; -0,75; 3, etc.). Direct input of fractions in symbolic form (e.g., 3/4, 5/2, etc.) is not supported.

Consequently, if the original model contains fractional coefficients, they must first be converted into decimal numbers or equivalently reformulated using integer coefficients.

Transformation of Constraints with Fractional Coefficients

Consider a constraint of the form:

$$\frac{p}{q}x_1 + \frac{r}{s}x_2 \leq \frac{t}{u},$$

where $p, q, r, s, t, u \in \mathbb{Z}$, with $q, s, u \neq 0$.

This constraint can be multiplied by a positive scalar m (for example, the least common multiple of q, s, u) in order to obtain an equivalent formulation with integer coefficients:

$$m\frac{p}{q}x_1 + m\frac{r}{s}x_2 \leq m\frac{t}{u}.$$

By choosing $m = \text{l.c.m.}(q, s, u)$, it is guaranteed that the resulting coefficients are integers, while preserving the equivalence of the original model.

It is important to emphasize that multiplying a constraint by a positive scalar does not alter the feasible region nor the set of optimal solutions (although it may affect the interpretation of certain quantities, such as the slack associated with that constraint).

Numerical Precision

Since the simulator operates using decimal representation, it is recommended to:

- Use a reasonable number of decimal digits.
- Avoid introducing excessively truncated approximations in models that are sensitive to small parametric variations.

In sensitivity analysis contexts, small decimal perturbations may induce structural changes in the optimal Basic Feasible Solution (BFS).

4. Visualization and Analysis Tools

In the graph panel (Solution Space), several controls are available that do not modify the formulated problem, but do enhance the understanding of its resolution::

-  **Labels:** displays labels at relevant points (BFSs, highlighted points, etc.). This facilitates the geometric interpretation of the feasible set.
-  **Fix Axes:** keeps the Cartesian axes scale constant. This is particularly useful in comparative analyses, as it prevents changes in the model from altering the visual scale.
-  **Isocost Lines:** allows the representation of the objective function level curves (different values of k). This helps to understand the direction of improvement toward optimality and to identify the optimal point or the set of optimal points.
-  **Dark Mode:** modifies the visualization scheme without altering the mathematical content.

-  **ES / EN / VAL:** allows switching the interface language and status messages without modifying the underlying mathematical model.

5. Guided Example: Formulation and Analysis

Consider the following problem:

Maximize:

$$z = 3x_1 + 2x_2$$

Subject to:

$$\begin{aligned}x_1 + x_2 &\leq 8 \\2x_1 + x_2 &\leq 10 \\x_2 &\leq 6 \\x_1, x_2 &\geq 0\end{aligned}$$

Procedure in the Simulator

1. In **Constraints**, click  until you have 3 constraints.
2. Enter the coefficients and the right-hand-side values:
 - R1: $1 \cdot x_1 + 1 \cdot x_2 \leq 8$
 - R2: $2 \cdot x_1 + 1 \cdot x_2 \leq 10$
 - R3: $0 \cdot x_1 + 1 \cdot x_2 \leq 6$
3. In **Objective Function**, enter:
 - *Max*
 - $c_1 = 3, c_2 = 2$
4. Activate  **Isocost Lines** and  **Fix Axes**.
5. Analyze the resulting feasible region.

The computed optimal solution corresponds to a Basic Feasible Solution (BFS), obtained at the intersection of two binding constraints.

 **Guiding question:** At which vertex (extreme point) of the convex polyhedron (feasible region, FR) is the optimum located? Which constraints intersect (are simultaneously binding) at that point?

6. Sensitivity Analysis

One of the main strengths of the simulator lies in its usefulness for parametric analysis.

6.1 Sensitivity with respect to the objective function coefficients c_j

Modifying c_1 or c_2 alters the slope of the level curves.

The following aspects should be analyzed:

- Whether the optimal solution remains at the same BFS.
- The range of values for which that BFS remains optimal (range of optimality).
- The change in the optimal objective function value z^* .

This behavior can be visually observed through the rotation of the level curves as the value of one or both c_j coefficients is modified.

6.2 Sensitivity with respect to the right-hand-side parameters b_i

Modifying b_i produces a parallel shift of the line associated with constraint i .

The following aspects should be analyzed:

- Whether the constraint is binding at the optimum.
- How the optimal solution and the optimal objective value z^* vary.
- Whether the modification alters the structure of the feasible set (feasible region).

This is directly related to the geometric interpretation of the **shadow price**.

7. Fundamental Concepts Illustrated with the Simulator

◆ **Feasible Region (FR).** Also referred to as the feasible set or region of feasibility, it is the set of all points (x_1, x_2) that simultaneously satisfy all the constraints of the problem, including the non-negativity constraints when imposed.

◆ **Basic Feasible Solutions (BFSs).** These are the extreme points (vertices) of the feasible region. In linear programming problems with two variables, BFSs correspond to the intersection between the boundary lines of two constraints.

◆ **Binding Constraints.** A constraint is binding at a BFS if it holds with equality at that solution. Geometrically, the BFS lies on the line associated with that constraint.

◆ **Slack.** For a constraint $a_1x_1 + a_2x_2 \leq b$, the slack (h) at the optimal point (x_1^*, x_2^*) is $h = b - (a_1x_1^* + a_2x_2^*)$. If it is positive, the constraint is non-binding. If it is equal to zero, the constraint is binding (i.e., it constitutes a bottleneck or limiting constraint).

◆ **Redundant Constraint.** A constraint is redundant if its removal does not modify the feasible region.

◆ **Bottleneck Constraint.** A constraint acts as a bottleneck when it is binding at the optimum; typically, small variations in its right-hand-side parameter will affect the optimal value of the objective function.

◆ **Multiple Optimal Solutions.** Graphically, these arise when the objective function line is parallel to an edge (boundary segment) of the feasible set and coincides with it at optimality. In this case, there exists an infinite set of optimal solutions along that segment.

8. How to Study with the Simulator

If you want to use it as a study tool, an effective method is:



1. **Build** a simple baseline **model** (2–3 constraints).
2. Activate  **Isocost Lines** and  **Fix Axes**.
3. **Change only one parameter** (c_j , b_i) **at a time**.
4. Analyze:
 - how the optimal solution and/or the objective function value changes,
 - which constraint is binding and which constraint has slack at the optimum,
 - when a constraint becomes redundant, and
 - when the problem becomes infeasible / unbounded / exhibits multiple optima.

In this way, you can use the simulator as a “laboratory” to analyze the model’s sensitivity.